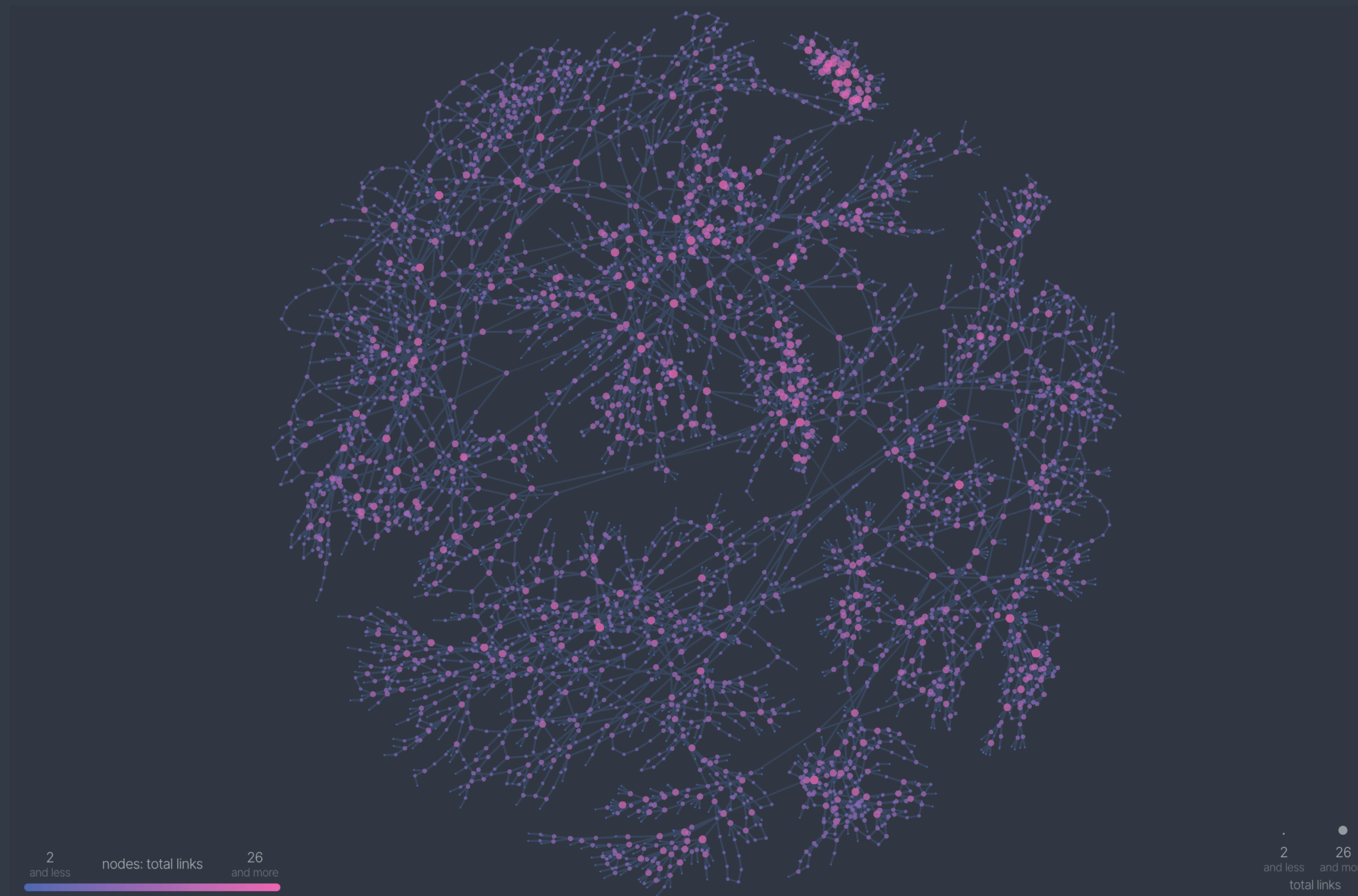


Resilience: cascades

Course: Physics of Complex Networks: Structure and Dynamics



US Power Grid

Objective: simulating cascade failures

(1) Write the code to simulate the Motter-Lai model

(2) Specifics for synthetic networks analysis

- Use $N = 5000$ (or larger); consider 20 instances of Erdos-Renyi and Barabasi-Albert with same $\langle k \rangle$
- Calculate how the relative size of the LCC change at the end of the cascade
- Compare: study the behavior for tuning the tolerance $\alpha \in [0,1]$
- How the result change for different seeds? Use: random, degree-rank, load-rank

Bonus: Watts threshold model for varying $\langle k \rangle$

(3) Empirical systems

- Simulate cascade failures for the US Power Grid
- What is the maximum tolerance above which the network gets disintegrated?
- How the result change if a Configuration Model (20 instances) is used instead?

